

Name-Hansika Kota

Role-AI/ML Intern

2D Virtual Try-On Research & Prototype

Started working on a basic 2D virtual try-on system to understand how clothing can be

Approach Followed

I started by taking a person image and a clothing image as inputs. The person image was processed using pose detection to identify body landmarks such as shoulders and upper body position. Based on the detected body points, garment dimensions were calculated and adjusted before placing the cloth on the person image. Image preprocessing was performed to prepare both images for overlay. Clothing alignment logic was applied to position garments according to body measurements and detected landmarks. Overlay blending was then used to generate the final try-on result.

For experimentation and execution support, I also explored **Google Cloud** services to understand how cloud platforms can support AI workflows, model execution, and scalable processing for virtual try-on systems.

Technologies Explored

- OpenCV — image preprocessing, resizing and garment overlay
- Media Pipe — pose detection and body landmark extraction
- TensorFlow — AI and virtual try-on research
- Human parsing concepts — body region understanding
- Google Cloud — explored for cloud-based execution and supporting AI experimentation

Research Findings

- Through this research, I understood the workflow of a 2D Virtual Try-On system including image preprocessing, pose detection, garment alignment, and clothing overlay. Using OpenCV and MediaPipe helped generate a basic try-on output. It was observed that body alignment improves garment placement, but realistic fitting requires advanced methods such as human parsing and VITON models. Google Cloud was also explored to support AI experimentation and model execution.



Code

```

C:\Users> Users > Hansika > Downloads > tryon.py > ...
1  import cv2
2  import mediapipe as mp
3  import numpy as np
4  from tkinter import Tk
5  from tkinter.filedialog import askopenfilename
6  Tk().withdraw()
7  print("Select PERSON image...")
8  person_path = askopenfilename(
9      title="Select Person Image",
10     filetypes=[("Image files", "*.jpg *.jpeg *.png")]
11 )
12 print("Select SHIRT/CLOTHING image...")
13 shirt_path = askopenfilename(
14     title="Select Shirt/Clothing Image",
15     filetypes=[("Image files", "*.jpg *.jpeg *.png")]
16 )
17 if not person_path or not shirt_path:
18     print("Please select both images.")
19     exit()
20 person = cv2.imread(person_path)
21 shirt = cv2.imread(shirt_path, cv2.IMREAD_UNCHANGED)
22 if person is None or shirt is None:
23     print("Could not load images.")
24     exit()
25 h, w, _ = person.shape
26 mp_pose = mp.solutions.pose
27 pose = mp_pose.Pose(
28     static_image_mode=True,
29     model_complexity=2,
30     min_detection_confidence=0.5
31 )
32 rgb = cv2.cvtColor(person, cv2.COLOR_BGR2RGB)
33 results = pose.process(rgb)
34 pose.close()
35 if not results.pose_landmarks:
36     print("No pose detected. Use a clear full body image.")
37     exit()

```

```

lm = results.pose_landmarks.landmark
l_shoulder = lm[mp_pose.PoseLandmark.LEFT_SHOULDER]
r_shoulder = lm[mp_pose.PoseLandmark.RIGHT_SHOULDER]
l_hip = lm[mp_pose.PoseLandmark.LEFT_HIP]
r_hip = lm[mp_pose.PoseLandmark.RIGHT_HIP]
ls_x = int(l_shoulder.x * w)
ls_y = int(l_shoulder.y * h)
rs_x = int(r_shoulder.x * w)
rs_y = int(r_shoulder.y * h)
lh_y = int(l_hip.y * h)
shirt_width = int(abs(ls_x - rs_x) * 2.0)
shirt_height = int(abs(lh_y - ls_y) * 1.2)
shirt_x = int((rs_x + ls_x) / 2) - shirt_width // 2
shirt_y = int((ls_y + rs_y) / 2) - int(shirt_height * 0.25)
shirt_x = max(0, shirt_x)
shirt_y = max(0, shirt_y)
if shirt_x + shirt_width > w:
    shirt_width = w - shirt_x
if shirt_y + shirt_height > h:
    shirt_height = h - shirt_y
shirt_resized = cv2.resize(shirt, (shirt_width, shirt_height))
output = person.copy()
if shirt_resized.shape[2] == 4:
    shirt_rgb = shirt_resized[:, :, :3]
    shirt_alpha = shirt_resized[:, :, 3] / 255.0
    for c in range(3):
        output[shirt_y:shirt_y + shirt_height,
               shirt_x:shirt_x + shirt_width, c] = (
            shirt_alpha * shirt_rgb[:, :, c] +
            (1 - shirt_alpha) * person[shirt_y:shirt_y + shirt_height,
                                       shirt_x:shirt_x + shirt_width, c]
        )
else:
    shirt_gray = cv2.cvtColor(shirt_resized, cv2.COLOR_BGR2GRAY)
    _, shirt_mask = cv2.threshold(shirt_gray, 240, 255, cv2.THRESH_BINARY_INV)
    shirt_mask = shirt_mask / 255.0
    for c in range(3):

```

```

else:
    shirt_gray = cv2.cvtColor(shirt_resized, cv2.COLOR_BGR2GRAY)
    _, shirt_mask = cv2.threshold(shirt_gray, 240, 255, cv2.THRESH_BINARY_INV)
    shirt_mask = shirt_mask / 255.0
    for c in range(3):
        output[shirt_y:shirt_y + shirt_height,
               shirt_x:shirt_x + shirt_width, c] = (
            shirt_mask * shirt_resized[:, :, c] +
            (1 - shirt_mask) * person[shirt_y:shirt_y + shirt_height,
                                       shirt_x:shirt_x + shirt_width, c]
        )
cv2.circle(output, (ls_x, ls_y), 6, (0, 255, 0), -1)
cv2.circle(output, (rs_x, rs_y), 6, (0, 255, 0), -1)
cv2.putText(output, "L.Shoulder", (ls_x + 5, ls_y - 5),
            cv2.FONT_HERSHEY_SIMPLEX, 0.4, (0, 255, 255), 1)
cv2.putText(output, "R.Shoulder", (rs_x + 5, rs_y - 5),
            cv2.FONT_HERSHEY_SIMPLEX, 0.4, (0, 255, 255), 1)
combined = cv2.hconcat([person, output])
cv2.putText(combined, "Original", (10, 30),
            cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 0, 255), 2)
cv2.putText(combined, "Try-On Output", (w + 10, 30),
            cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 0, 255), 2)
cv2.imwrite("tryon_output.jpg", combined)
cv2.imshow("Virtual Try-On Demo", combined)
cv2.waitKey(0)
cv2.destroyAllWindows()
print("Done! Output saved as tryon_output.jpg")

```

2 - 3D Virtual Try-On Research

For this task, I explored how 3D virtual try-on systems work and how clothes can be fitted digitally instead of using simple image overlay. I studied concepts like avatar creation, body mesh generation, and cloth simulation to understand how realistic fitting is achieved.

I also explored tools like Unity, Blender, Three.js and SMPL to understand how 3D rendering and virtual fitting work. Google Cloud was also explored to support AI experimentation and execution.

Explore Tools

- Explored Unity to understand real-time rendering and interactive virtual experiences.
- Studied Blender to learn 3D avatar and garment visualization.
- Explored Three.js for browser-based 3D rendering concepts.
- Researched SMPL models to understand body mesh generation.
- Explored AR frameworks to understand real-time virtual fitting experiences.

Research Findings

I found that 3D try-on gives a more realistic fitting experience because it considers body shape and cloth behaviour instead of directly placing clothes on images. It requires more processing but provides better visualization and fitting accuracy.

3- AI Model Exploration

Human Pose Estimation

Studied how pose estimation models detect body key points and understand human posture for accurate garment positioning in virtual try-on systems.

Body Landmark Detection

Explored body landmark detection to identify important body regions such as shoulders, arms, and upper body for better clothing alignment.

AI Fashion Recommendation Integration

Researched how AI can be integrated to suggest suitable clothing based on style, appearance, and user preference for a better try-on experience.

Segmentation Models

Studied segmentation techniques to separate body and clothing regions for improving fitting accuracy and overlay quality.

Models Explored

- MediaPipe Pose — explored for body landmark and pose detection.
- Detectron2 — studied for segmentation and object understanding.
- U2Net — explored for background and region separation.
- OpenPose — researched for multi-point human pose detection.

4-Dataset Collection Task

Work Done

For this task, I explored datasets that can support virtual try-on development and understood how different datasets contribute to clothing understanding, pose detection, and body segmentation.

Dataset Name	Source	Use Case
DeepFashion Dataset	DeepFashion	Used for fashion clothing images and garment understanding
MPII Human Pose Dataset	MPII	Used for human pose estimation and landmark detection
CIHP Dataset	CIHP	Used for body segmentation and human parsing
VITON Dataset	VITON	Used for apparel try-on and clothing fitting
ModaNet Dataset	ModaNet	Used for clothing detection and segmentation

Research Findings

Through dataset exploration, I understood that different datasets support different stages of virtual try-on systems such as clothing understanding, pose estimation, body segmentation, and realistic garment fitting.